

AI and Large Language Models: A Deeper Dive into Grok

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Abstract

Generative AI and language models have been in the spotlight for the past few years for many reasons. Whether it be the stock market and the exploding financial market of AI or the many uses of using it for making work more efficient and seamless or everyday tasks like drafting emails or using AI as a tool, it's. AI has also made its way into other apps we use in our everyday life. The newest apple update has AI features to turn on your do not disturb and have ai to decipher important messages to bypass the do not disturb and alert you to important messages (Apple, 2025a). As well as search engines like google and safari having built in ai's in their search engine. This paper will also discuss LLMs and explore Grok as an AI in specific. Discussing its fact-checking and use cases and benefits as well as the mistakes and errors ai can make as well as any biases that may occur in AI.

What is an LLM and How Does It Work?

Large Language Models (LLMs) are massive AI brains built to understand, generate, and manipulate human language (Amazon Web Services, n.d.). They are trained on an absurd amount of text, as well as data from forums, books, news dumps, any sources the LLM can get correct data from. The main processes that take place happen via neural networks. They are algorithms that mimic how our brains connect the dots and figure things out. Transformers are architectures that focus on word relationships, weighing what is important in a sentence (IBM, n.d.).

For example, if you were to search "Paris is the capital of..." Before completing the sentence, the option of "France" will appear this is because it is seen that pattern many times. The model calculates the likelihoods for the next word or phrase. They smooth out these processes for specific gigs, like chatting casually or generating code snippets. As well as evolving and learning with feedback loops from users, getting better over time.

LLMs are not fully thinking they are pattern recognizing machines That means they can output incorrect information, inventing facts that sound legit but are not true. Early models like GPT-2 would ramble incoherently if pushed too far. Now with versions like Grok-3 that have trillions of tokens in training, they are smoother, but still prone to glitches. Overall, LLMs are incredibly important to the AI space, powering everything from auto-complete in emails to full-blown story generators.

Describe One AI Chatbot/LLM You Explored

I decided to explore one of the AI's I have enjoyed using lately for the past few months. It is Grok. Xai was released in late 2023 (xAI, n.d.). One of the main differences and tools I love about grok is the Realtime pulls from X (formerly Twitter). When using X, you can use grok to verify fact check any false claims or ask the AI to explain the tweet. The interface using it in app is also very seamless as there is a grok button on all tweets on X (Gupta, n.d.).

Grok-1, the original version, started off as an open-source AI with 314 billion parameters, using a Mixture of Experts setup where specialized sub models worked together to complete tasks. Then Grok-2 was focused on increasing efficiency, better at multilingual tasks and following and listening to instructions. Now, Grok-3's the latest version of grok currently using 12.8 trillion tokens during training on a fleet of Nvidia GPUs (Hyperright, n.d.; LifeArchitect.ai, n.d.). It handles text chats, creates images with its Aurora engine, and can aid in coding or voice modes on apps.

Grok is an incredibly accessible AI there is a free tier on x.com or the official grok website Premium tiers like X Premium+ or Super Grok unlock features like unlimited chats. Grok is integrated with X Which gives Grok the ability and opportunity to reply and engage with tweets. Elon Musk the creator of grok describes the chatbot as "maximum truth-seeking," but controversies and scandals like the anti-Semitic comments grok was outputting for a brief period towards Jewish people call some of groks objectivity into this case (Al Jazeera, 2025; NPR, 2025).

My User Experience with Grok

The setup and usage of grok is incredibly simple you can either simply open X click on the grok tab at the bottom of the page and start typing. You do not need to do any other sign up or logging in, you can simply access grok with your existing X account. I first used grok like any other AI simply asking it basic questions and fact-finding sort of questions. The main purpose and reason I prefer grok to other AI's is the constant updating of information which leads to and allows for being able to fact check and ask Ai about current events. The downside of grok is the emits bite after 10-15 solid queries, it slows you down, forcing waits or lighter use.

There are many Use cases like for example. In brainstorming mode, it can be highly effective for creating study sheets sorting through lots of basic history facts or notes for school. Grok was able to go through notes I have and make concise and easy to remember cheat sheets to study off from and saved me lots of time. Another notable example is groks ability to scan through code and search for errors that may be causing code to create errors and can suggest fixes and corrections.

As what I alluded to earlier that have been use cases in which grok has provided and sent out incorrect information due to bugs one of these times was when grok was engaging in spitting out antisemitic comments even without being prompted to (CNN Business, 2025). As well as there was a period in which grok was providing incorrect information about south African genocide. These use cases of the Ai not doing what it has intended has been rare on a personal level for me

I have a pleasant experience for the most part but there are times when grok makes egregious bad mistakes.

Do I Fact-Check Everything from Grok? Why or Why Not?

No Fact checking every AI claim would take a long time. I only fact check if I am using grok for an important topic or if I feel as though the AI has said something wrong about a topic I know about, so I continue to test the ai to see if it will correct itself. Grok also provides the sources it is using so if I ever feel skeptical about the information or want to read it directly, I will click on the links and go through them and see where grok is getting its information from. When using grok AI for tasks like work or studying or learning new information and skills it's important to check and see where the information you're getting is good when grok first came out I was skeptical of it because of Elon Musks connection and the political implications of him possibly biasing the AI but after engaging with it for some time now the mistakes it has made have been signification but overall it has been more factual than not (Learn Prompting, n.d.).

Two Scenarios for Using Grok to Boost Learning in This Course

Scenario one: A group project that a team is working on and I may forget about a topic like integrating LLMs into business models. You can ask Grok a prompt like "explain real-world LLM use in marketing, pros and cons." It provides examples like personalized ads via sentiment analysis with X trends for currency. I would then be able to refine my section and add those insights with real data and information quickly.

Scenario two: I upload my notes and ask Grok to help me create a comprehensive cheat sheet and knowledge based study sheet I can expand the prompt add in things that are specific I want grok to focus on and go more in depth well as continuing to probe and ask grok for any questions I may be unsure about.

Conclusion

Grok and other AI tools are great and fun to learn and use and can be great at assisting and helping your day-to-day life. It is important to study and understand these tools the processes behind them and understanding how they work and how we can use them to our best advantage

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